

Measurement-Efficient Port-Reading Policies for Fluid Antenna Systems Under Spatially Correlated κ - μ Shadowed Fading

Matheus F. Silva, Pedro M. R. Pereira, Felipe A. P. de Figueiredo, and Rausley A. A. de Souza

Abstract—Fluid antenna systems exploit spatial diversity by selecting among multiple candidate ports, but practical receivers may be unable to read all ports before selection. This work studies the port-reading problem, namely, which ports should be measured under a strict observation budget. Under spatially correlated κ - μ shadowed fading, geometry-driven uniform reading, threshold-aware marginal ranking, and greedy coverage are compared in terms of observed-only outage probability. For an aperture with 100 candidate ports and observation budgets of 3 to 7 measured ports, the results show a clear budget-dependent behavior. Greedy coverage is most effective with three reads, achieving the lowest outage in 12 of 14 fading regimes. For four or more measured ports, uniform subsampling becomes the most reliable strategy and achieves the best average outage across regimes. Marginal ranking consistently underperforms, indicating that per-port success frequency is a poor proxy for set-level diversity in correlated apertures. These results establish port reading as a critical design layer for measurement-efficient operation of fluid antennas.

Keywords—Fluid antenna systems, port selection, port reading, outage probability, spatial correlation, κ - μ shadowed fading, Monte Carlo.

I. INTRODUCTION

Fluid antenna systems (FASs) represent a promising antenna technology for beyond-5G and 6G systems, especially when terminal size and hardware constraints limit conventional multi-antenna arrays [1]. By electronically moving one radiating element among multiple candidate positions, a FAS exploits spatial diversity with a single radio frequency (RF) chain, making it attractive for compact terminals and sensor-class devices [2, 3]. When the best port is selected, this diversity gain can approach or even exceed that of classical combining schemes despite the small aperture [4]. Nevertheless, such gains often require reading many ports, which increases measurement overhead, switching delay, and receiver complexity.

Partial-observation FAS operation mitigates this burden by measuring only a subset of ports before the final selection decision. Prior studies have considered machine learning (ML)-based port selection, deep-learning transceiver strategies, and generative methods that infer unobserved ports from sparse measurements [5-9]. These works mainly emphasize the selector or predictor applied after measurements are acquired. However, when only a few ports can be read, the sensing pattern itself may dominate the final outage performance.

This observation reveals an underexplored design layer in partial-observation FAS. Spatial correlation makes port reading

nontrivial because individually promising ports may provide redundant information when measured together. Therefore, a design-stage evaluation of the port-reading problem is needed before adding reconstruction or learning-based selection. Such an evaluation clarifies whether the measured set should prioritize geometric coverage, marginal port quality, or complementarity among ports.

This work treats port reading as a policy-level design problem for measurement-limited FAS operation. The evaluation isolates the effect of the measured set from downstream reconstruction by using an observed-only outage metric, in which the receiver selects only among the measured ports. Under spatially correlated κ - μ shadowed fading, three low-complexity policies are compared: uniform geometric reading, threshold-aware marginal ranking, and greedy coverage. The results show a budget-dependent crossover. Greedy coverage is preferable for three measured ports before selection, whereas uniform reading is more robust when at least four ports are measured. Marginal ranking is consistently inferior, showing that isolated per-port quality is not a reliable surrogate for set-level diversity in correlated apertures.

The main contributions are as follows. First, port reading is formulated as a policy-level design problem for measurement-limited FAS, separating measured-port choice from downstream reconstruction or selection. Second, an observed-only outage metric isolates the effect of the measured set under identical budgets and fading conditions. Third, uniform reading, marginal ranking, and greedy coverage are compared under spatially correlated κ - μ shadowed fading. Fourth, a Monte Carlo evaluation with 100 candidate ports, 14 fading regimes, and budgets $\{3, 4, 5, 6, 7\}$ measured ports shows that greedy coverage is best for 3 measured ports, whereas uniform reading is more robust when at least 4 ports are measured. Finally, the results show that per-port exceedance frequency is a weak diversity proxy, highlighting the need for measured-set-level reading policies.

II. DATASET AND SYSTEM MODEL

A. Signal Model and Partial Observation

A downlink scenario is considered, as illustrated in Fig. 1, in which a single-antenna base station (BS) serves a terminal equipped with an FAS composed of M candidate ports distributed over a normalized electrical aperture $D = L/\lambda$. Here, L is the physical aperture length and λ is the carrier wavelength. Hence, D is dimensionless and represents the aperture size measured in wavelengths. Let $u \in \mathbb{C}$ denote the transmitted symbol with average power $\mathbb{E}[|u|^2] = P_u$. The

The authors are with the National Institute of Telecommunications (Inatel), Santa Rita do Sapucaí, MG - 37536-00, Brazil. E-mails: {matheus.f, pedro.marcio, felipe.figueiredo, rausley}@inatel.br

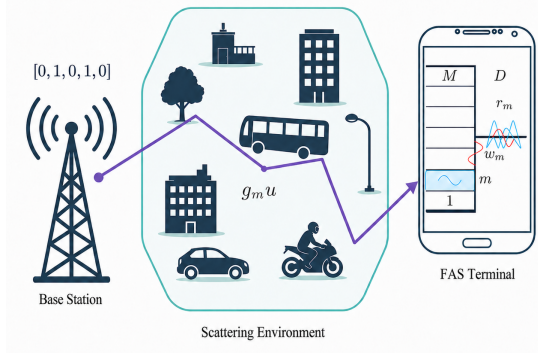


Fig. 1. Downlink FAS model.

baseband sample collected at port $m \in \{1, \dots, M\}$ is modeled as

$$r_m = g_m u + w_m, \quad (1)$$

where M denotes the total number of available FAS ports, g_m is the complex channel coefficient associated with port m , and $w_m \sim \mathcal{CN}(0, N_0)$ is circularly symmetric complex Gaussian noise. The instantaneous signal-to-noise ratio (SNR) at port m is therefore

$$\eta_m = \frac{P_u}{N_0} |g_m|^2, \quad (2)$$

and the full SNR profile is collected as $\boldsymbol{\eta} = [\eta_1, \dots, \eta_M]^T$.

To reduce measurement overhead, only a subset of ports is observed. Let $\mathcal{O}$ denote the index set of measured ports with $|\mathcal{O}| = N \ll M$, i.e., N denotes the number of measured ports, and define the observation vector $\boldsymbol{\eta}_{\mathcal{O}} = [\eta_{m_1}, \dots, \eta_{m_N}]^T$ for $\mathcal{O} = \{m_1, \dots, m_N\}$. Under partial observation, any observed-only selector or profile-reconstruction method must operate from $\boldsymbol{\eta}_{\mathcal{O}}$ rather than from the full profile $\boldsymbol{\eta}$.

B. Spatially Correlated κ - μ Shadowed Fading

The ports are assumed to experience spatial correlation due to sub-wavelength sampling along the aperture. For isotropic 2D scattering in the azimuthal plane, the correlation between ports m and n is modeled through a Jakes-type matrix $\mathbf{R} \in \mathbb{C}^{M \times M}$ with entries

$$[\mathbf{R}]_{m,n} = J_0 \left(\frac{2\pi|m-n|}{D(M-1)} \right), \quad (3)$$

where $J_0(\cdot)$ denotes the zeroth-order Bessel function of the first kind. Following [9], let the eigendecomposition be $\mathbf{R} = \mathbf{V}\boldsymbol{\Sigma}\mathbf{V}^H$, and define the correlation-shaping matrix $\mathbf{B} = \mathbf{V}\boldsymbol{\Sigma}^{1/2}$.

To capture multipath clustering and shadowing of the dominant component, the port-wise channel power is generated by adapting the physical κ - μ shadowed construction in [10, eq. (7)] to the multiport FAS setting. For port m , the received power is modeled as

$$|g_m|^2 = \sum_{\ell=1}^{\mu} |q_{m,\ell} + p_m|^2, \quad (4)$$

where $q_{m,\ell}$ denotes the scattered component of cluster ℓ at port m , and p_m denotes the shadowed dominant component at that port. Spatial correlation is then introduced across ports by generating the scattered vectors as $\mathbf{q}_{\ell} = \sigma_q \mathbf{B} \mathbf{z}_{\ell}$, where

$\mathbf{q}_{\ell} = [q_{1,\ell}, \dots, q_{M,\ell}]^T$, $\mathbf{z}_{\ell} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_M)$, and σ_q^2 controls the average scattered power.

The dominant vector is $\mathbf{p} = \xi \mathbf{v}$, where $\mathbf{v} \in \mathbb{C}^M$ has identical entries v_0 , and ξ is a Nakagami- m random variable with $\mathbb{E}[\xi^2] = 1$. Hence, the dominant component is perfectly correlated across ports because the same shadowed amplitude and deterministic entry are applied to all positions. This restrictive assumption makes the dominant term a common aperture component, while spatial diversity is mainly produced by the scattered vectors shaped by $\mathbf{B}$. Since all ports are identically distributed, the dominant-to-scattered ratio is set by $|v_0|^2 = \kappa \sigma_q^2$, equivalently $\kappa = |v_0|^2 / \sigma_q^2$. Thus, increasing κ strengthens a common dominant contribution, whereas decreasing κ makes port-to-port differences depend more strongly on the correlated scattered components. The parameters κ , μ , and m retain their standard meanings as dominant-to-scattered power ratio, number of clusters, and shadowing severity, respectively [11].

C. Port-Selection Metric

With full observation, the ideal FAS selects the port that maximizes the instantaneous SNR,

$$\eta^* = \max_{1 \leq m \leq M} \eta_m. \quad (5)$$

Given a threshold η_{th} , the outage probability (OP) is defined as

$$P_{\text{out}} = \Pr(\eta^* < \eta_{\text{th}}), \quad (6)$$

and is approximated through Monte Carlo evaluation over independent channel realizations.

Under partial observation, the selector maximizes only over the measured ports,

$$\eta^{\text{obs}} = \max_{m \in \mathcal{O}} \eta_m, \quad (7)$$

yielding $P_{\text{out}}^{\text{obs}} = \Pr(\eta^{\text{obs}} < \eta_{\text{th}})$. This metric isolates the port-reading policy from any reconstruction model. A reconstruction-aided receiver could instead estimate $\hat{\boldsymbol{\eta}}$ from $\boldsymbol{\eta}_{\mathcal{O}}$ and select $\hat{m} = \arg \max_m \hat{\eta}_m$, but this stage is not optimized here because the objective is to evaluate the measured set itself.

D. Dataset Construction

A multi-fading dataset is generated to promote robustness across heterogeneous conditions. For each parameter set (κ, μ, m) in Table I, independent realizations of the complete SNR profile $\boldsymbol{\eta}$ are produced using M candidate ports distributed over the normalized electrical aperture D [12]. For each realization, the observed index set $\mathcal{O}$ is generated according to a port-reading policy and an observation budget N , where $N \ll M$. The complete set of parameter combinations is summarized in Table I¹.

¹For numerical stability, the values 0.001 and 50 denote limiting cases tending to 0 and ∞ , respectively.

TABLE I
 FADING MODEL PARAMETERS USED IN THE DATASET.

Dataset	κ	μ	m	Channel distribution
1	0.001	1	0.001	Rayleigh, severe shadowing
2	0.001	1	2	Rayleigh, moderate shadowing
3	0.001	1	50	Rayleigh
4	0.001	2	50	Nakagami
5	0.001	5	50	Nakagami
6	5	1	0.001	Rician, severe shadowing
7	5	1	2	Rician, moderate shadowing
8	5	1	50	Rician
9	5	2	0.001	κ - μ , severe shadowing
10	5	2	2	κ - μ , moderate shadowing
11	5	2	50	κ - μ
12	5	5	0.001	κ - μ , severe shadowing
13	5	5	2	κ - μ , moderate shadowing
14	5	5	50	κ - μ

III. PORT-READING POLICIES AND EVALUATION FRAMEWORK

A. Reconstruction Context and Observed-Only Evaluation

Under partial observation, a receiver may reconstruct $\eta \in \mathbb{R}^M$ from $\eta_{\mathcal{O}} \in \mathbb{R}^N$ before selection through

$$\hat{\eta} = f(\eta_{\mathcal{O}}), \quad (8)$$

where f may be a parametric ML regressor or model-based interpolator. In supervised learning, f is commonly trained by minimizing a reconstruction loss, such as

$$\mathcal{L}_{\text{MSE}} = \frac{1}{M} \sum_{m=1}^M (\hat{\eta}_m - \eta_m)^2. \quad (9)$$

Although f determines reconstruction fidelity, $\mathcal{O}$ determines which information reaches the input and constrains the final decision. Therefore, this work uses observed-only outage to isolate the reading policy from any specific reconstruction model. The results should be interpreted as a controlled design-stage comparison of measurement patterns, not as an optimized reconstruction-aided receiver.

B. Port-Reading Strategies

Port reading selects a subset of ports to probe under a fixed budget. Three low-complexity policies are compared under identical budgets. Policies B and C produce ordered port lists, whereas Policy A produces a deterministic geometry-driven subset for each budget.

Policy A uniformly subsamples the aperture by selecting N indices evenly spaced between ports 1 and M . It prioritizes spatial coverage and requires no data-driven tuning. Policy B is a threshold-aware marginal ranking learned from a design subset $\mathcal{D}_{\text{des}}$. With exceedance indicator $b_{i,m} = \mathbf{1}\{\eta_m^{(i)} > \eta_{\text{th}}\}$, it scores each port by

$$\hat{p}_m = \frac{1}{|\mathcal{D}_{\text{des}}|} \sum_{i \in \mathcal{D}_{\text{des}}} b_{i,m}, \quad (10)$$

and ranks ports by decreasing $\hat{p}_m$. Ties are resolved by design-set mean SNR and then by port index, favoring ports that individually exceed the threshold most often.

Policy C is a greedy coverage policy that accounts for measured-set complementarity. Starting from an empty set, it

Algorithm 1: MONTE CARLO EVALUATION OF PORT-READING POLICIES

Require: $\mathcal{D} = \{\eta^{(i)}\}_{i=1}^S$, η_{th} , $\mathcal{N}$, R , ρ

```

1 for each regime do
2   for  $r = 1, \dots, R$  do
3     split  $\mathcal{D}$  into  $\mathcal{D}_{\text{des}}$  and  $\mathcal{D}_{\text{eval}}$ , with fractions  $1 - \rho$  and  $\rho$ ,
4     respectively; compute  $b_{i,m} = \mathbf{1}\{\eta_m^{(i)} > \eta_{\text{th}}\}$  on  $\mathcal{D}_{\text{des}}$ 
5     learn Policy B by sorting  $\hat{p}_m$ , and learn Policy C by greedy
6     coverage with Policy B tie-breaks
7     for each  $N \in \mathcal{N}$  do
8       construct  $\mathcal{O}_A(N)$ ,  $\mathcal{O}_B(N)$ , and  $\mathcal{O}_C(N)$ 
9       evaluate observed-only outage for Policies A–C and ideal
10      all-port outage on  $\mathcal{D}_{\text{eval}}$ 
11
12 aggregate per-budget means and variability summaries
    
```

adds one port per step to maximally reduce observed-only outage on the design split. At step k , with selected set $\mathcal{S}_{k-1}$ and uncovered samples $\mathcal{U}_{k-1}$, the next port is

$$m^* = \arg \max_{m \notin \mathcal{S}_{k-1}} \sum_{i \in \mathcal{U}_{k-1}} b_{i,m}, \quad (11)$$

which maximizes the number of newly covered samples. If all design samples are covered before the order is complete, the remaining positions are filled using the Policy B ranking over unselected ports, preserving determinism and the same threshold-aware fallback criterion.

C. Per-Regime Monte Carlo Evaluation Protocol

Each policy is evaluated independently in each fading regime using repeated-split Monte Carlo simulation. Let $S = |\mathcal{D}|$, R , ρ , and $\mathcal{N}$ denote the number of channel samples, split repetitions, evaluation fraction, and budget set. In each repetition, samples are deterministically permuted and split into $\mathcal{D}_{\text{des}}$ and $\mathcal{D}_{\text{eval}}$, with $|\mathcal{D}_{\text{eval}}| = \rho S$ and $|\mathcal{D}_{\text{des}}| = (1 - \rho)S$. Policies B and C are learned on $\mathcal{D}_{\text{des}}$ and evaluated on disjoint $\mathcal{D}_{\text{eval}}$, while Policy A is evaluated on the same samples for paired comparisons.

For a read set $\mathcal{O}$, the primary metric is

$$\hat{P}_{\text{out}}^{\text{obs}}(\mathcal{O}) = \frac{1}{|\mathcal{D}_{\text{eval}}|} \sum_{i \in \mathcal{D}_{\text{eval}}} \mathbf{1}\left\{\max_{m \in \mathcal{O}} \eta_m^{(i)} < \eta_{\text{th}}\right\}. \quad (12)$$

The ideal baseline uses all M ports. Since Eq. (12) depends only on exceedance events, the evaluation may equivalently use a binary threshold matrix, preserving the metric while simplifying Policy C. Uncertainty is measured from repetition-to-repetition variability. Per-budget outage values are averaged across R repetitions, and Policies A and C are compared by

$$\Delta_r(N) = \hat{P}_{\text{out}}^{\text{obs}}(\mathcal{C}; r, N) - \hat{P}_{\text{out}}^{\text{obs}}(\mathcal{A}; r, N), \quad (13)$$

computed within each repetition on the same evaluation subset. Algorithm 1 summarizes the procedure.

IV. RESULTS AND DISCUSSION

The Monte Carlo evaluation follows Section III-C with $S = 10^6$ samples per fading regime, $M = 100$ ports, $D = 1$, $R = 100$, $\rho = 0.2$, and $\mathcal{N} = \{3, 4, 5, 6, 7\}$. Each split therefore uses 8×10^5 design and 2×10^5 evaluation samples. The threshold is set to $\eta_{\text{th}} = 0.8$, slightly below the normalized average SNR,

TABLE II

PERCENTAGE OF REGIMES WON BY EACH POLICY AT EACH BUDGET.					
Policy	$N = 3$ (%)	$N = 4$ (%)	$N = 5$ (%)	$N = 6$ (%)	$N = 7$ (%)
A	14.3%	85.7%	92.9%	100.0%	92.9%
B	0.0%	0.0%	0.0%	0.0%	0.0%
C	85.7%	14.3%	7.1%	0.0%	7.1%

Fig. 2. Best port-reading policy by fading regime and observation budget.

which avoids degenerate low-threshold conditions in which most ports succeed and high-threshold conditions in which outages dominate. Thus, policy differences remain visible, and each repetition reports observed-only outage for Policies A–C and the ideal all-port baseline. For reproducibility, the implementation code is publicly available².

Table II and Fig. 2 summarize the regime-wise best policies. Policy C is strongest at the tightest budget, achieving the lowest outage in 12 of 14 regimes at $N = 3$. For $N \geq 4$, Policy A becomes dominant and is the unique winner at $N = 6$. Policy B never achieves the lowest mean outage. Fig. 3 confirms the same trend through the percentage of evaluation samples for which the observed set contains at least one port above threshold, $100 \times (1 - \hat{P}_{\text{out}}^{\text{obs}})$. Policy A has the highest mean above-threshold rate for $N \geq 4$, while Policy C remains strongest at $N = 3$.

Fig. 4 shows the crossover between Policies A and C using the per-dataset paired mean outage difference for $N = 3$ and $N = 4$. Differences are mostly negative at $N = 3$, favoring Policy C, and mostly positive at $N = 4$, favoring Policy A once one outlier dataset is read. Fig. 5 shows a representative regime. All policies improve as N increases because additional measured ports increase the chance of observing a port above threshold. However, Policies A and C approach the ideal baseline much faster than Policy B, confirming that empirical per-port exceedance frequency is insufficient in strongly correlated and statistically homogeneous apertures. Thus, complementarity of the measured set is more relevant than isolated port quality.

Threshold sensitivity was assessed by repeating learning and testing for $\eta_{\text{th}} \in \{0.6, 0.7, 0.8, 0.9, 1.0\}$, with Policies B and C recomputed for each threshold because both depend on exceedance events. Larger η_{th} increases the mean observed

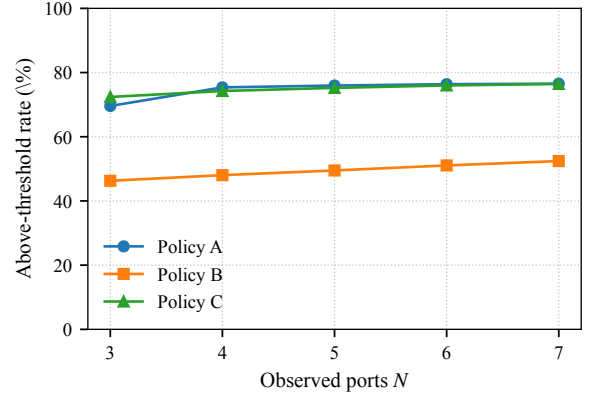
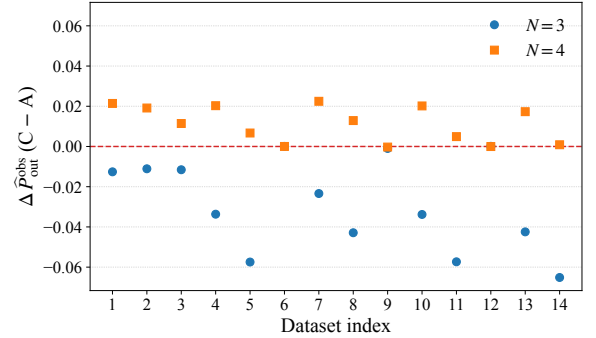


Fig. 3. Mean percentage of evaluation samples above threshold across all regimes.


 Fig. 4. Forest plot of per-dataset paired mean differences in observed-only outage between Policies C and A for $N = 3$ and $N = 4$. Negative values favor Policy C.

outage for all policies, but the relative behavior remains stable. Policies A and C remain close and substantially outperform Policy B. Relative to $\eta_{\text{th}} = 0.8$, policy classification changes by only 10.0%, 5.7%, 7.1%, and 14.3% at $\eta_{\text{th}} = 0.6, 0.7, 0.9$, and 1.0 , respectively. Hence, the conclusion is not tied to a single threshold: uniform reading is robust across the tested operating range, while greedy coverage is most beneficial in the most measurement-limited case.

The shared dominant-component assumption helps explain the policy comparison. When $\kappa \approx 0$, the channel is scattered-field dominated, and port-to-port diversity is mainly governed by the spatial correlation matrix. When $\kappa = 5$, the dominant component is stronger but common to all ports, so it does not create additional spatial diversity by itself. It instead shifts the aperture channel level coherently, leaving the scattered components as the main source of relative port differences. Therefore, marginal ranking is weak because ports have similar marginal statistics, and isolated exceedance frequency does not reliably identify complementary subsets. Uniform reading samples scattered spatial variations across the aperture, while greedy coverage exploits complementarity when only $N = 3$ ports are read. For $N \geq 4$, uniform coverage captures enough scattered diversity while avoiding data-driven over-specialization to the design split.

Stability was assessed using 95% intervals from repeated splits. Regime-specific intervals are computed across $R = 100$ splits, whereas the aggregate above-threshold-rate curve in Fig. 3 first averages repetitions within each regime and then computes variability across the 14 regimes. In Fig. 5, split variability is much smaller than inter-policy differences. At

²<https://github.com/MatheusFS-dev/fas-reading-policy-benchmark>

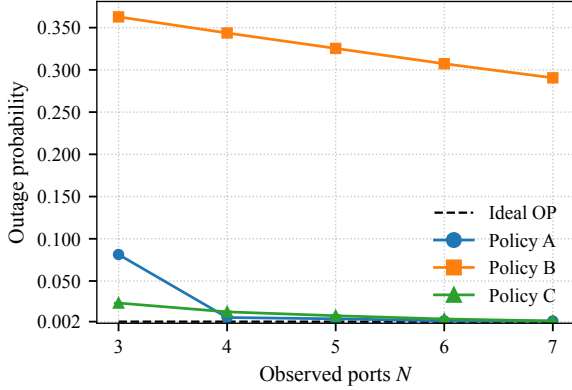


Fig. 5. Observed-only outage versus number of observed ports in the $\kappa \rightarrow 0$, $\mu = 5$, $m \rightarrow \infty$ regime.

$N = 3$, Policy C obtains 0.0239 ± 0.0001 outage, while Policy A obtains 0.0814 ± 0.0001 . At $N = 4$, the ordering reverses: Policy A obtains 0.0068 , with interval half-width below 10^{-4} , whereas Policy C obtains 0.0135 ± 0.0001 . Thus, random split variability does not explain the crossover.

The empirical selection-time benchmark further supports the practical relevance of the crossover. Policy A required 0.0638 ms on average, with 95th percentile 0.0703 ms, whereas Policy C required 1595.63 ms on average, with 95th percentile 2020.06 ms. Thus, Policy C is approximately 2.50×10^4 times slower in terms of mean selection time. Although the greedy order can be computed offline, this overhead matters when policies are recomputed for different regimes, thresholds, or deployment conditions. Therefore, for $N \geq 4$, Policy A combines strong outage performance with substantially lower selection cost.

Overall, port reading should be treated as a design layer rather than as a secondary implementation detail. Greedy coverage is useful in the extremely measurement-limited regime because it explicitly seeks complementary ports. Once a modest number of ports can be observed, however, uniform spatial sampling becomes a robust and simple default. This conclusion is reinforced by the selection-time benchmark, since Policy A remains competitive for $N \geq 4$ while avoiding the substantial data-driven cost of Policy C.

V. CONCLUSION

This work studied port reading as a distinct design layer in measurement-limited fluid antenna system (FAS) operation. A controlled observed-only outage benchmark isolated the measured port set from downstream reconstruction. Under spatially correlated κ - μ shadowed fading with $M = 100$ ports, repeated-split Monte Carlo evaluation across 14 regimes showed that the best reading strategy depends strongly on the measurement budget.

Complementarity-aware greedy coverage provides the best outage protection when only three ports can be read. With four or more ports, uniform reading becomes the most robust default across the evaluated regimes. Policy-construction-time measurements further show that Policy A is orders of magnitude faster to construct than Policy C, supporting its use when the budget is not extremely tight. By contrast, threshold-aware marginal ranking is consistently weak, confirming that

per-port exceedance frequency is not a reliable proxy for set-level diversity in correlated apertures. These results are specific to the considered aperture, threshold, and spatial-correlation model. In particular, the dominant component was modeled as perfectly correlated across ports, making the scattered field the main source of spatial diversity.

Future work could extend the framework to different aperture lengths, such as $D = 0.5$ and $D = 2$, to quantify how stronger or weaker spatial correlation changes the relative behavior of the reading policies. Another relevant extension is correlation-aware subset selection, for example, choosing ports that maximize the determinant of the corresponding channel-covariance submatrix. Such a criterion is attractive because it directly promotes spatial diversity and could compete with data-driven policies without requiring training samples. Further extensions include port-dependent dominant components, reconstruction-aided selection, adaptive online port reading, and joint sensing-policy optimization under practical switching, latency, and hardware constraints.

ACKNOWLEDGMENTS

This work was partially supported by CNPq (306199/2025-4), by the project XGM-AFCCT-2024-9-1-1 supported by xGMobile - EMBRAPII-Inatel Competence Center on 5G and 6G Networks, with financial resources from the PPI IoT/Manufatura 4.0 from MCTI, 052/2023, signed with EMBRAPII, by RNP Grant No. 01245.020548/2021-07, under the Brazil 6G project of the Radiocommunication Reference Center (CRR) of the National Institute of Telecommunications (Inatel), by FAPEMIG (PPE-00124-23, RED-00194-23, APQ-04523-23, APQ-05305-23 and APQ-03162-24), and by FINEP (No 1060/2 contract 01.25.0883.00).

REFERENCES

- [1] K.-K. Wong, K.-F. Tong *et al.*, “Bruce Lee-inspired fluid antenna system: Six research topics and the potentials for 6G,” *Front. Commun. Netw.*, vol. 3, p. 853416, 2022.
- [2] K.-K. Wong, A. Shojafard *et al.*, “Fluid antenna systems,” *IEEE Trans. Wireless Commun.*, vol. 20, no. 3, pp. 1950–1962, Mar. 2021.
- [3] K.-K. Wong and K.-F. Tong, “Fluid antenna multiple access,” *IEEE Trans. Wireless Commun.*, vol. 21, no. 7, pp. 4801–4815, 2021.
- [4] X. Lai, T. Wu *et al.*, “On the performance of fluid antenna system using maximum ratio combining,” *IEEE Commun. Lett.*, vol. 28, no. 2, pp. 402–406, 2024.
- [5] Z. Chai, K.-K. Wong *et al.*, “Port selection for fluid antenna systems,” *IEEE Commun. Lett.*, vol. 26, no. 5, pp. 1180–1184, 2022.
- [6] M. Eskandari, A. G. Burr *et al.*, “cGAN-based slow fluid antenna multiple access,” *IEEE Wireless Commun. Lett.*, vol. 13, no. 10, pp. 2907–2911, 2024.
- [7] F. S. dos S. Silveira, A. Flávia dos Reis *et al.*, “Comparison of multi-label classification and regression models for port selection in fluid antenna systems,” in *Proc. 2025 IEEE Latin Conference on IoT (LCIoT)*, 2025, pp. 198–201.
- [8] P. D. Alvim, H. S. Silva *et al.*, “LNN-powered fluid antenna multiple access,” *arXiv preprint arXiv:2507.08821*, 2025.
- [9] N. Waqar, K.-K. Wong *et al.*, “Deep learning enabled slow fluid antenna multiple access,” *IEEE Commun. Lett.*, vol. 27, no. 3, pp. 861–865, 2023.
- [10] L. Moreno-Pozas, F. Lopez-Martinez *et al.*, “The κ - μ shadowed fading model: Unifying the κ - μ and η - μ distributions,” *IEEE Trans. Veh. Technol.*, vol. 65, no. 12, pp. 9630–9641, 2016.
- [11] J. F. Paris, “Statistical characterization of κ - μ shadowed fading,” *IEEE Trans. Veh. Technol.*, vol. 63, no. 2, pp. 518–526, 2014.
- [12] M. F. Silva, P. M. R. Pereira *et al.*, “Low-power AI-based port selection for fluid antennas in spatially correlated channels,” in *Proc. 2025 Workshop on Communication Networks and Power Systems (WCNPS)*, 2025.